

2021 AIME II Problems/Problem 13

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Problem

Find the least positive integer n for which $2^n + 5^n - n$ is a multiple of 1000.

Solution 1

Recall that 1000 divides this expression if 8 and 125 both divide it. It should be fairly obvious that $n \geq 3$; so we may break up the initial condition into two sub-conditions.

(1) $5^n \equiv n \pmod{8}$. Notice that the square of any odd integer is 1 modulo 8 (proof by plugging in $1^2, 3^2, 5^2, 7^2$ into modulo 8), so the LHS of this expression goes $5, 1, 5, 1, \dots$, while the RHS goes $1, 2, 3, 4, 5, 6, 7, 8, 1, \dots$. The cycle length of the LHS is 2, RHS is 8, so the cycle length of the solution is $\text{lcm}(2, 8) = 8$. Indeed, the n that solve this congruence are exactly those such that $n \equiv 5 \pmod{8}$.

(2) $2^n \equiv n \pmod{125}$. This is extremely computationally intensive if we try to calculate all $2^1, 2^2, \dots, 2^{100} \pmod{125}$, so we take a divide-and-conquer approach instead. In order for this expression to be true, $2^n \equiv n \pmod{5}$ is necessary; it shouldn't take too long for us to go through the 20 possible LHS-RHS combinations, considering that n must be odd. We only need to test 10 values of n and obtain $n \equiv 3 \pmod{20}$ or $n \equiv 17 \pmod{20}$.

With this in mind we consider $2^n \equiv n \pmod{25}$. By the Generalized Fermat's Little Theorem, $2^{20} \equiv 1 \pmod{25}$, but we already have n modulo 20. Our calculation is greatly simplified. The LHS's cycle length is 20 and the RHS's cycle length is 25, from which their least common multiple is 100. In this step we need to test all the numbers between 1 to 100 that $n \equiv 3 \pmod{20}$ or $n \equiv 17 \pmod{20}$. In the case that $n \equiv 3 \pmod{20}$, the RHS goes $3, 23, 43, 63, 83$, and we need $2^n \equiv n \equiv 2^3 \pmod{25}$; clearly $n \equiv 83 \pmod{100}$. In the case that $n \equiv 17 \pmod{20}$, by a similar argument, $n \equiv 97 \pmod{100}$.

In the final step, we need to calculate 2^{97} and 2^{83} modulo 125:

Note that $2^{97} \equiv 2^{-3}$; because $8 \cdot 47 = 376 \equiv 1 \pmod{125}$, we get $2^{97} \equiv 47 \pmod{125}$.

Note that 2^{83} is $2^{-17} = 2^{-16} \cdot 2^{-1}$. We have

$$2^{-1} \equiv 63,$$

$$2^{-2} \equiv 63^2 \equiv 2000 \equiv 21$$

$$\begin{aligned}
2^{-1} &\equiv 03^{-1} = 3909 \equiv -31, \\
2^{-4} &\equiv (-31)^2 = 961 \equiv -39, \\
2^{-8} &\equiv 1521 \equiv 21, \\
2^{-16} &\equiv 441, \\
2^{-17} &\equiv 63 \cdot 441 \equiv 7 \cdot (-31) = -217 \equiv 33.
\end{aligned}$$

This time, LHS cycle is 100, RHS cycle is 125, so we need to figure out n modulo 500. It should be $n \equiv 283, 297 \pmod{500}$.

Put everything together. By the second subcondition, the only candidates less than 1000 are 283, 297, 783, 797. Apply the first subcondition, $n = \boxed{797}$ is the desired answer.

~Ross Gao (Solution)

~MRENTHUSIASM (Minor Reformatting)

Solution 2

We have that $2^n + 5^n \equiv n \pmod{1000}$, or $2^n + 5^n \equiv n \pmod{8}$ and $2^n + 5^n \equiv n \pmod{125}$ by CRT. It is easy to check $n < 3$ don't work, so we have that $n \geq 3$. Then, $2^n \equiv 0 \pmod{8}$ and $5^n \equiv 0 \pmod{125}$, so we just have $5^n \equiv n \pmod{8}$ and $2^n \equiv n \pmod{125}$. Let us consider both of these congruences separately.

First, we look at $5^n \equiv n \pmod{8}$. By Euler's Totient Theorem (ETT), we have $5^4 \equiv 1 \pmod{8}$, so $5^5 \equiv 5 \pmod{8}$. On the RHS of the congruence, the possible values of n are all nonnegative integers less than 8 and on the RHS the only possible values are 5 and 1. However, for 5^n to be 1 $\pmod{8}$ we must have $n \equiv 0 \pmod{4}$, a contradiction. So, the only possible values of n are when $n \equiv 5 \pmod{8} \implies n = 8k + 5$.

Now we look at $2^n \equiv n \pmod{125}$. Plugging in $n = 8k + 5$, we get $2^{8k+5} \equiv 8k + 5 \pmod{125} \implies 2^{8k} \cdot 32 \equiv 8k + 5 \pmod{125}$. Note, for $2^n \equiv n \pmod{125}$ to be satisfied, we must have $2^n \equiv n \pmod{5}$ and $2^n \equiv n \pmod{25}$. Since $2^{8k} \equiv 1 \pmod{5}$ as $8k \equiv 0 \pmod{4}$, we have $2 \equiv -2k \pmod{5} \implies k = 5m - 1$. Then, $n = 8(5m - 1) + 5 = 40m - 3$. Now, we get $2^{40m-3} \equiv 40m - 3 \pmod{125}$. Using the fact that $2^n \equiv n \pmod{25}$, we get $2^{-3} \equiv 15m - 3 \pmod{25}$. The inverse of 2 modulo 25 is obviously 13, so $2^{-3} \equiv 13^3 \equiv 22 \pmod{25}$, so $15m \equiv 0 \pmod{25} \implies m = 5s$. Plugging in $m = 5s$, we get $n = 200s - 3$.

Now, we are finally ready to plug n into the congruence modulo 125. Plugging in, we get $2^{200s-3} \equiv 200s - 3 \pmod{125}$. By ETT, we get $2^{100} \equiv 1 \pmod{125}$, so $2^{200s-3} \equiv 2^{-3} \equiv 47 \pmod{125}$. Then, $200s \equiv 50 \pmod{125} \implies s \equiv 4 \pmod{5} \implies s = 5y + 4$. Plugging this in, we get $n = 200(5y + 4) - 3 = 1000y + 797$, implying the smallest value of n is simply $\boxed{797}$.

~rocketsri

Solution 3 (Chinese Remainder Theorem and Binomial Theorem)

We wish to find the least positive integer n for which $2^n + 5^n - n \equiv 0 \pmod{1000}$. Rearranging gives

$$2^n + 5^n \equiv n \pmod{1000}.$$

Applying the Chinese Remainder Theorem, we get the following system of congruences:

$$\begin{aligned}2^n + 5^n &\equiv n \pmod{8}, \\2^n + 5^n &\equiv n \pmod{125}.\end{aligned}$$

It is clear that $n \geq 3$, from which we simplify to

$$5^n \equiv n \pmod{8}, \tag{1}$$

$$2^n \equiv n \pmod{125}. \tag{2}$$

We solve each congruence separately:

- For (1), quick inspections produce that $5^1, 5^2, 5^3, 5^4, \dots$ are congruent to $5, 1, 5, 1, \dots$ modulo 8, respectively. More generally, $5^n \equiv 5 \pmod{8}$ if n is odd, and $5^n \equiv 1 \pmod{8}$ if n is even. As 5^n is always odd (so is n), we must have $n \equiv 5 \pmod{8}$.

That is, $n = 8r + 5$ for some nonnegative integer r .

- For (2), we substitute the result from (1) and simplify:

$$\begin{aligned}2^{8r+5} &\equiv 8r + 5 \pmod{125} \\(2^8)^r \cdot 2^5 &\equiv 8r + 5 \pmod{125} \\256^r \cdot 32 &\equiv 8r + 5 \pmod{125} \\6^r \cdot 32 &\equiv 8r + 5 \pmod{125}.\end{aligned}$$

Note that $5^3 = 125$ and $6 = 5 + 1$, so we apply the Binomial Theorem to the left side:

$$\begin{aligned}(5+1)^r \cdot 32 &\equiv 8r + 5 \pmod{125} \\ \left[\binom{r}{0} 5^0 + \binom{r}{1} 5^1 + \binom{r}{2} 5^2 + \underbrace{\binom{r}{3} 5^3 + \dots + \binom{r}{r} 5^r}_{0 \pmod{125}} \right] \cdot 32 &\equiv 8r + 5 \pmod{125} \\ \left[1 + 5r + \frac{25r(r-1)}{2} \right] \cdot 32 &\equiv 8r + 5 \pmod{125} \\ 32 + 160r + 400r(r-1) &\equiv 8r + 5 \pmod{125} \\ 32 + 35r + 25r(r-1) &\equiv 8r + 5 \pmod{125} \\ 25r^2 + 2r + 27 &\equiv 0 \pmod{125}. \tag{*}\end{aligned}$$

Since $125 \equiv 0 \pmod{5}$, it follows that

$$\begin{aligned}25r^2 + 2r + 27 &\equiv 0 \pmod{5} \\2r + 2 &\equiv 0 \pmod{5} \\r &\equiv 4 \pmod{5}.\end{aligned}$$

That is, $r = 5s + 4$ for some nonnegative integer s .

Substituting this back into (*), we get

$$\begin{aligned}25(5s+4)^2 + 2(5s+4) + 27 &\equiv 0 \pmod{125} \\625s^2 + 1010s + 435 &\equiv 0 \pmod{125} \\10s + 60 &\equiv 0 \pmod{125} \\10(s+6) &\equiv 0 \pmod{125} \\s + 6 &\equiv 0 \pmod{25}\end{aligned}$$

$$5 + 5 = 10 \pmod{25}.$$

Therefore, the least such nonnegative integer s is 19.

Finally, combining the bolded results from above generates the least such positive integer n at $s = 19$:

$$\begin{aligned} n &= 8r + 5 \\ &= 8(5s + 4) + 5 \\ &= 40s + 37 \\ &= \boxed{797}. \end{aligned}$$

~MRENTHUSIASM (inspired by Math Jams's **2021 AIME II Discussion**)

Solution 4 (Minimal Computation)

$$5^n \equiv n \pmod{8}$$

Note that $5^n \equiv 5, 1, 5, 1, \dots$ and $n \equiv 0, \dots, 7$ so n is periodic every $[2, 8] = 8$. Easy to check that only $n \equiv 5 \pmod{8}$ satisfy. Let $n = 8a + 5$. Note that by binomial theorem,

$$2^{8a+5} = 32 \cdot 2^{8a} \equiv 7(1 + 15)^{2a} \equiv 7(1 + 30a) \pmod{25}$$

So we have

$$7(1 + 30a) \equiv 8a + 5 \pmod{25} \implies 202a \equiv 23 \implies 2a \equiv 23 + 25 \implies a \equiv 24 \pmod{25}$$

. Combining $a \equiv 24 \pmod{25}$ with $n \equiv 5 \pmod{8}$ gives that n is in the form of $197 + 200k$,

$n = 197, 397, 597, 797$. Note that since $\phi(125) = 100$

$$2^{197+200k} \equiv 2^{197} \equiv 2^{-3} \equiv \underbrace{47}_{\text{Extended Euclidean Algorithm}} \pmod{125}$$

Easy to check that only $\boxed{797} \equiv 47 \pmod{125}, 1797 \equiv 47 \pmod{125}, 2797 \equiv 47 \pmod{125}, \dots$

~Afo

Solution 5 (STEP by step transform)

1. The desired n has the form $n = m + 20l + 100p$, where m, l, p are integers and $m < 20$.

Really:

$$(2^{m+20} - (m + 20)) - (2^m - m) = (2^{m+20} - 2^m) - 20.$$

The first term is a multiple of 100 (**Claim**).

We denote **step** an increase in m by 20. At each **step**, the divisibility by 10 is preserved, the number of tens is reduced by 2.

$$(2^{m+100} - (m + 100)) - (2^m - m) = (2^{m+100} - 2^m) - 100.$$

We denote **STEP** an increase in m by 100. At each **STEP**, the first term is a multiple of 1000, which means that at each **STEP** the divisibility by 100 is preserved, the number of hundreds decreases by 1.

2. If the expression $2^n + 5^n - n$ is a multiple of 1000, the number n is odd (2^n is even, 5^n is odd), which means that 5^n ends in 125. Therefore, the number $2^n - n$ ends in 875.

3. $2^3 - 3 = 8 - 3 = 5$. The tens digit is even (0), it cannot be transformed into 7 in several steps, since at each **step** this digit changes by 2.

$17 = 20 - 3$, so $2^{17} + 2^3$ is a multiple of 10, $(2^{17} - 17) + (2^3 - 3) = 2^{17} + 2^3 - 20$ is a multiple of 10.

$$2^{17} \pmod{100} = ((2^{10} \pmod{100}) \cdot (2^7 \pmod{100})) \pmod{100} = (24 \cdot 28) \pmod{100} = 72.$$

$$(2^{17} - 17) \pmod{100} = 55.$$

The tens digit 5 is odd, subtracting 2 at each **step** in 4 **steps** of 20 will turn it into 7. So

$$(2^{97} - 97) \pmod{100} = 75.$$

$$2^{97} \pmod{1000} = ((2^{49} \pmod{1000}) \cdot 2^7) \pmod{1000} = 672.$$

$$(2^{97} - 97) \pmod{1000} = 575.$$

We transform 5 into 8 in 7 **STEPS**, so

$$(2^{797} - 797) \pmod{1000} = 875.$$

$$(5^{797} + 2^{797} - 797) \pmod{1000} = 0.$$

Note, that for $n = 797 + 1000k$, k is an integer, the expression $2^n + 5^n - n$ is a multiple of 1000.

Claim

The numbers $2^{n+2 \cdot 5^k} + 2^n = 2^n(2^{2 \cdot 5^k} + 1)$ and $2^{n+4 \cdot 5^k} - 2^n = 2^n(2^{4 \cdot 5^k} - 1)$

are a multiple of 10^{k+1} for any $n > k$, where n, k are an integer.

Proof

First, if $m \pmod{10} \equiv 4$, then $(m^4 - m^3 + m^2 - m + 1) \pmod{10} \equiv 6 - 4 + 6 - 4 + 1 = 5$.

$2^{4n+2} \pmod{10} \equiv 4$. So, if the number $2^{4n+2} + 1$ is a multiple of 5^k , then $2^{20n+10} + 1$ is a multiple of 5^{k+1} .

Really, denote $m = 2^{4n+2}$, then, $2^{20n+10} + 1 = m^5 + 1 = (m + 1) \cdot (m^4 - m^3 + m^2 - m + 1)$.

The first factor is a multiple of 5^k , the second factor is a multiple of 5. Their product is a multiple of 5^{k+1} .

For odd n , using Newton's binomial for 4^n , we get $2^{2n} + 1$ is a multiple of 5.

$$2^{2n} + 1 = 4^n + 1 = (5 - 1)^n + 1 = 5^n - n \cdot 5^{n-1} + \dots + 5n$$

If $n = 5^k$, then $n > k$, each term is a multiple of $5n$.

Therefore, the number $4^{5^k} + 1$ is a multiple of 5^{k+1} and $2^{k+1}(2^{2 \cdot 5^k} + 1)$ is a multiple of 10^{k+1} .

The difference

$$2^{n+4 \cdot 5^k} - 2^n = 2^n(2^{4 \cdot 5^k} - 1) = 2^n(2^{2 \cdot 5^k} - 1)(2^{2 \cdot 5^k} + 1)$$

is a multiple of 10^{k+1} .

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Solution 6 (Art of Enumeration)

Problem require us to find minimum n where $2^n + 5^n - n$ is a multiple of 1000.

Since we already know that multiplication does not affect mod, we can simply just focus on enumerating and hope to find some pattern.

The pattern of 5^n is pretty simple, it starts with 5, 25, and then repeat the pattern 125, 625, in other words,
 $5^n \equiv 125 \pmod{1000}$ when n is odd, and $5^n \equiv 625 \pmod{1000}$ when n is even if we ignore $n = 1$ and $n = 2$.

The pattern of 2^n , however, is quite long. In fact, it happens when $n = 103$, where $2^{103} \equiv 008 \pmod{1000}$. More specifically, the pattern starts with $n = 3$, which means for every 100 n , $2^n \pmod{1000}$ will repeat in a loop:

008, 016, 032, 064, 128, 256, 512, 024, 048, 096, 192, 384, 768, 536, 072, 144, 288, 576, 152, 304, 608, 216, 432, 864, 728, 456, 912, 824, 648, 296, 592, 184, 368, 736, 472, 944, 888, 776, 552, 104, 208, 416, 832, 664, 328, 656, 312, 624, 248, 496, 992, 984, 968, 936, 872, 744, 488, 976, 952, 904, 808, 616, 232, 464, 928, 856, 712, 424, 848, 696, 392, 784, 568, 136, 272, 544, 088, 176, 352, 704, 408, 816, 632, 264, 528, 056, 112, 224, 448, 896, 792, 584, 168, 336, 672, 344, 688, 376, 752, 504

Moreover, we might found that for every 20 consecutive integer, $2^n \pmod{100}$ will repeat in a loop. Thus, by adding 5^n to first twenty element, which is $n = 3$ to $n = 22$, we can get a pattern of last two digit of $2^n + 5^n \pmod{1000}$, which is

33, 41, 57, 89, 53,
 81, 37, 49, 73, 21,
 17, 09, 93, 61, 97,
 69, 13, 01, 77, 29

By subtracting n , we can also get following sequence:

30, 37, 52, 83, 46,
 73, 28, 39, 62, 09,
 04, 95, 78, 45, 80,
 51, 94, 81, 56, 07

Noticing that only when $n = 3$ and $n = 17$, $2^n + 5^n - n \equiv 0 \pmod{10}$, so $n = 3 + 20k$ and $17 + 20k$ are our possible candidates. However, since the length of our pattern is 20, we know that $n = 3 + 20k$ is definitely not answer, which only left us $n = 17 + 20k$.

Now, we calculate the last three digit of $2^n + 5^n - n$ at $n = 17, n = 37, n = 57, n = 77, n = 97$, which is 180, 560, 940, 320, 700.

Since the pattern will repeat for every 100 consecutive integer, the only thing that could possibly change last three digit is $-n$, noticing that only when $n = 97$, $2^n + 5^n - n \equiv 0 \pmod{100}$. So the answer only could be $n = 97 + 100k_2$, by subtracting 700, which means adding 700 to n , we can get our final answer $n = \boxed{797}$.

Also, by this method we can ensure that $\frac{2^n + 5^n - n}{1000}$ is an integer only if $n = 797 + 1000k$, all the calculation can be done without a calculator (about 15 minutes or so), need some patience though.

If you are wondering, $2^{103} = 10,141,204,801,825,835,211,973,625,643,008$.

~henry_in_out

Note

If you are wondering, $\frac{2^{797} + 5^{797} - 797}{1000}$ is equal to the following 555-digit number:

119,975,745,111,650,476,385,411,555,010,245,228,283,196,935,699,128,663,834,
 986,755,809,416,868,468,175,224,221,677,450,271,793,186,899,399,171,810,240,
 468,879,630,691,320,210,517,618,332,910,064,430,017,422,410,572,497,874,327,
 116,662,273,049,144,209,532,072,513,248,821,826,125,454,909,131,367,242,561,
 087,064,439,357,641,814,942,784,478,465,638,163,997,895,630,266,780,260,171,
 070,180,244,384,687,160,174,154,618,815,254,719,975,476,350,789,415,969,908,
 281,131,044,034,581,864,573,596,962,567,993,948,922,431,587,652,735,290,158,
 293,666,986,616,531,419,688,663,485,548,648,995,509,247,893,796,528,146,109,
 144,143,744,650,882,271,050,771,309,301,498,467,898,226,649,089,924,520,539,
 820,344,035,886,481,987,499,589,845,734,228,834,491,187.

Video Solution

2021 AIME II #13

[MathProblemSolvingSkills.com](https://www.mathproblemsolving.com)

Video Solution by Interstigation

Did not use any advanced method, easily understandable:

<https://youtu.be/ThF67J8iaS0>

~Interstigation

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